Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

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7F Consequences of Singular Value Decomposition

Exercise 7F1

Prove that if $S, T \in \mathcal{L}(V, W)$, then $\big|\|S\| - \|T\|\big| \le \|S - T\|$.

Axler Note. *The inequality above is called the* reverse triangle inequality.

Solution. By 7.87(d), $\|S\| = \|(S-T)+T\| \le \|S-T\| + \|T\|$. Similarly, $\|T\| = \|(T-S)+S\| \le \|T-S\| + \|S\|$. By 7.87(c) with $\lambda = -1$, $\|T-S\| = \|-(S-T)\| = \|S-T\|$. Combining the inequalities yields the desired result. ■

Exercise 7F2 ★

Suppose that $T \in \mathcal{L}(V)$ is self-adjoint or that $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that

$$\|T\| = \max\{|\lambda| \mid \lambda \text{ is an eigenvalue of } T\}.$$

Solution. If $V = \{0\}$, then T has no eigenvalues, so we adopt the convention that the maximum over the empty set is 0. In this case, the only operator on V is $T = 0$, so $\|T\| = 0$ by 7.87(b), and the equality holds. Now assume $V \neq \{0\}$. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of T, where each eigenvalue λ of T appears in the list as many times as $\dim E(\lambda, T)$. As in the solution to Exercise 7E7, $n = \dim V \ge 1$. By Exercise 7E7, the singular values of T are $|\lambda_1|, \dots, |\lambda_n|$ sorted in decreasing order. Each eigenvalue λ of T appears in the list $\lambda_1, \dots, \lambda_n$ at least once since $\dim E(\lambda, T) \ge 1$. We then have

$$\max_{1 \le k \le n} |\lambda_k| = \max\{|\lambda| \mid \lambda \text{ is an eigenvalue of } T\}.$$

By 7.88(a), $\|T\|$ is the largest singular value of T, which is the left-hand side of the equality in the previous line. ■

Exercise 7F3

Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. Prove that

$$\|Tv\| = \|T\|\,\|v\| \iff T^*Tv = \|T\|^2 v.$$

Solution.

$\boxed{\Rightarrow}$ Define $S = \|T\|^2 I - T^*T$. By 7.5, T^*T is self-adjoint. Since $\|T\|^2 \in \mathbb{R}$, $(\|T\|^2 I)^* = \|T\|^2 I$. Hence, S is self-adjoint. For every $u \in V$, 7.88(c) gives

$$\langle Su, u\rangle = \langle(\|T\|^2 I - T^*T)u, u\rangle = \|T\|^2\|u\|^2 - \|Tu\|^2 \ge 0.$$

Hence, S is a positive operator. By hypothesis, $\langle Sv, v\rangle = \langle(\|T\|^2 I - T^*T)v, v\rangle = \|T\|^2\|v\|^2 - \|Tv\|^2 = 0$. Since S is positive, $\langle Sv, v\rangle = 0$ implies $Sv = 0$ by 7.43. Therefore, $(\|T\|^2 I - T^*T)v = 0$, which gives $T^*Tv = \|T\|^2 v$.

$\boxed{\Leftarrow}$ Suppose $T^*Tv = \|T\|^2 v$. Then $\|Tv\|^2 = \langle Tv, Tv\rangle = \langle T^*Tv, v\rangle = \|T\|^2\langle v, v\rangle = \|T\|^2\|v\|^2$. Taking square roots gives $\|Tv\| = \|T\|\,\|v\|$. ■

Exercise 7F4

Suppose $T \in \mathcal{L}(V, W)$, $v \in V$, and $\|Tv\| = \|T\| \, \|v\|$. Prove that if $u \in V$ and $\langle u, v \rangle = 0$, then $\langle Tu, Tv \rangle = 0$.

Solution. Suppose $u \in V$ with $\langle u, v \rangle = 0$. By Exercise 7F3, $T^*Tv = \|T\|^2 v$. By the definition of the adjoint, the previous line, and 6.6(e), since $\|T\|^2 \in \mathbb{R}$, we have

$$\langle Tu, Tv \rangle = \langle u, T^*Tv \rangle = \langle u, \|T\|^2 v \rangle = \|T\|^2 \langle u, v \rangle = 0. \qquad \blacksquare$$

Exercise 7F5

Suppose U is a finite-dimensional inner product space, $T \in \mathcal{L}(V, U)$, and $S \in \mathcal{L}(U, W)$. Prove that

$$\|ST\| \leq \|S\| \, \|T\|.$$

Solution. By 7.87(a), $\|S\| \geq 0$. By 7.88(c) applied to S and T, for every $v \in V$, we have

$$\|STv\| \leq \|S\| \, \|Tv\| \leq \|S\| \, \|T\| \, \|v\|.$$

By 7.88(c), $\|ST\|$ is the smallest number such that $\|STv\| \leq \|ST\| \, \|v\|$ for all $v \in V$. Hence, $\|ST\| \leq \|S\| \, \|T\|$. $\blacksquare$

Exercise 7F6

Prove or give a counterexample: If $S, T \in \mathcal{L}(V)$, then $\|ST\| = \|TS\|$.

Solution. The statement is false. Suppose $\dim V \geq 2$. By 6.35, there exists an orthonormal basis $e_1, e_2, \ldots, e_{\dim V}$ of V. Define the maps S and T by

$$Sv = \langle v, e_1 \rangle e_1, \quad Tv = \langle v, e_1 \rangle e_2$$

for all $v \in V$. By 6.6(a), $S, T \in \mathcal{L}(V)$. Then $STv = \langle v, e_1 \rangle Se_2 = \langle v, e_1 \rangle \langle e_2, e_1 \rangle e_1 = 0$ for all $v \in V$. On the other hand, $TSv = \langle v, e_1 \rangle Te_1 = \langle v, e_1 \rangle e_2 = Tv$ for all $v \in V$. Therefore, $ST = 0$ and $TS = T$. Moreover, $T \neq 0$ because $Te_1 = e_2 \neq 0$. By 7.87(a) and (b), we have $\|ST\| = 0 < \|TS\| = \|T\|$. $\blacksquare$

Exercise 7F7

Show that defining $d(S, T) = \|S - T\|$ for $S, T \in \mathcal{L}(V, W)$ makes d a metric on $\mathcal{L}(V, W)$.
Axler Note. *This exercise is intended for readers who are familiar with metric spaces.*

Solution. We verify the metric axioms for d:

- **Nonnegativity and identity**: Nonnegativity follows from 7.87(a). By 7.87(b), $d(S, T) = \|S - T\| = 0$ if and only if $S - T = 0$, i.e., $S = T$.
- **Symmetry**: Since $T - S = (-1)(S - T)$, 7.87(c) implies $\|T - S\| = \|(-1)(S - T)\| = \|S - T\|$, so $d(S, T) = d(T, S)$.

- **Triangle inequality**: For any $R \in \mathcal{L}(V,W)$, $S-R=(S-T)+(T-R)$, so by 7.87(d),

$$d(S,R)=\|S-R\| \leq \|S-T\|+\|T-R\|=d(S,T)+d(T,R).$$

Therefore, $(\mathcal{L}(V,W),d)$ is a metric space. ■

Exercise 7F8

(a) Prove that if $T \in \mathcal{L}(V)$ and $\|I-T\|<1$, then T is invertible.
(b) Suppose that $S \in \mathcal{L}(V)$ is invertible. Prove that if $T \in \mathcal{L}(V)$ and $\|S-T\|<1/\|S^{-1}\|$, then T is invertible.

Axler Note. *This exercise shows that the set of invertible operators in $\mathcal{L}(V)$ is an open subset of $\mathcal{L}(V)$, using the metric defined in* Exercise 7F7.

Solution.

(a) By 3.65, it suffices to show that T is injective. Let $v \in V$ be such that $Tv=0$. Then $(I-T)v=v$. By 7.88(c), $\|I-T\|\,\|v\| \geq \|(I-T)v\|=\|v\|$. If $v \neq 0$, then dividing both sides by $\|v\|$ gives $\|I-T\| \geq 1$, which contradicts the assumption that $\|I-T\|<1$. Therefore, $v=0$, and T is injective.
(b) Note that the hypothesis $\|S-T\|<1/\|S^{-1}\|$ requires that $\|S^{-1}\| \neq 0$, so $\|S^{-1}\|>0$ by 7.87(a). By 3.8, $I-S^{-1}T=S^{-1}(S-T)$. By Exercise 7F5,

$$\|I-S^{-1}T\| \leq \|S^{-1}\|\,\|S-T\|.$$

Since $\|S-T\|<1/\|S^{-1}\|$ and $\|S^{-1}\|>0$, $\|I-S^{-1}T\|<1$. By part (a), $S^{-1}T$ is invertible, so $T=S(S^{-1}T)$ is invertible. ■

Exercise 7F9

Suppose $T \in \mathcal{L}(V)$. Prove that for every $\varepsilon>0$, there exists an invertible operator $S \in \mathcal{L}(V)$ such that $0<\|T-S\|<\varepsilon$.

Manual Remark. *Note that this exercise assumes $V \neq \{0\}$, for otherwise, $\|T-S\|=0$ for every $S \in \mathcal{L}(V)$.*

Solution. Since $(0,\varepsilon)$ contains infinitely many real numbers, and T has at most $\dim V$ distinct eigenvalues by 5.12, there exists a real number $\lambda \in (0,\varepsilon)$ that is not an eigenvalue of T. Consider the operator $S=T-\lambda I$. By 5.7, S is invertible because λ is not an eigenvalue of T. Moreover, since I is an isometry, its singular values are all 1 by 7.69, so $\|I\|=1$ by 7.88(a). Therefore, by 7.87(c), $\|T-S\|=\|\lambda I\|=|\lambda|\|I\|=\lambda$, where $|\lambda|=\lambda$ because $\lambda>0$. Hence, $0<\|T-S\|=\lambda<\varepsilon$. ■

Exercise 7F10

Suppose $\dim V>1$ and $T \in \mathcal{L}(V)$ is not invertible. Prove that for every $\varepsilon>0$, there exists $S \in \mathcal{L}(V)$ such that $0<\|T-S\|<\varepsilon$ and S is not invertible.

Solution. By 3.65, T is not injective since it is not invertible. Then there exists a nonzero $u \in \operatorname{null} T$. Let $U=(\operatorname{span}(u))^{\perp}$. Since $\dim \operatorname{span}(u)=1$ and $\dim V>1$, we have $\dim U=\dim V-1 \geq 1$ by 6.51.

Consider the operator $S \in \mathcal{L}(V)$ defined by $S = T + \frac{\varepsilon}{2} P_U$. Note that for every $w \in U$, $\langle w, u \rangle = 0$, so $u \in U^\perp$ and hence $P_U u = 0$ by 6.57(c). Then $Su = 0$, so S is not injective, and hence not invertible by 3.65. By 6.57(h), $\|P_U v\| \leq \|v\|$ for all $v \in V$, which yields $\|P_U\| \leq 1$ by 7.88(c). Since $\dim U \geq 1$, there exists a unit vector $x \in U$. By 6.57(b), $P_U x = x$, so $\|P_U\| \geq \|P_U x\| = \|x\| = 1$ by 7.88(b). Thus, $\|P_U\| = 1$. Finally, by 7.87(c),

$$\|T - S\| = \left\| -\frac{\varepsilon}{2} P_U \right\| = \frac{\varepsilon}{2} \|P_U\| = \frac{\varepsilon}{2},$$

so $0 < \|T - S\| < \varepsilon$. ■

Exercise 7F11

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that for every $\varepsilon > 0$ there exists a diagonalizable operator $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \varepsilon$.

Manual Remark. *Note that this exercise assumes $V \neq \{0\}$, for otherwise, $\|T - S\| = 0$ for every $S \in \mathcal{L}(V)$.*

Solution. Let $\dim V = n$. By Schur's theorem, since $\mathbb{F} = \mathbb{C}$, there exists an orthonormal basis $e_1, \ldots, e_n$ of V such that $\mathcal{M}(T)$ is upper triangular with respect to this basis with diagonal entries $\lambda_1, \ldots, \lambda_n$, which are the eigenvalues of T by 5.41, possibly with repetitions. Define $\delta_1 = \frac{\varepsilon}{2}$. For $k = 2, \ldots, n$, choose $\delta_k \in [0, \frac{\varepsilon}{2})$ such that

$$\lambda_k + \delta_k \notin \{\lambda_j + \delta_j \mid 1 \leq j < k\}.$$

Note that this construction is possible since $[0, \frac{\varepsilon}{2})$ contains infinitely many real numbers, and only $k - 1$ values are excluded for each $k \geq 2$. By the linear map lemma, we may construct an operator $D \in \mathcal{L}(V)$ by $De_k = \delta_k e_k$. Then $\mathcal{M}(D, (e_1, \ldots, e_n))$ is diagonal with diagonal entries $\delta_1, \ldots, \delta_n$, which are its eigenvalues by 5.41.

Let $S = T + D$. By 3.35, $\mathcal{M}(S, (e_1, \ldots, e_n))$ is upper triangular with diagonal entries $\lambda_1 + \delta_1, \ldots, \lambda_n + \delta_n$, which are distinct by construction. Since these are n distinct eigenvalues by 5.41, S is diagonalizable by 5.58. By the complex spectral theorem, D is normal since it has a diagonal matrix with respect to the orthonormal basis $e_1, \ldots, e_n$. By Exercise 7F2, and since $\delta_k \geq 0$ for all k,

$$\|T - S\| = \|D\| = \max_{1 \leq k \leq n} |\delta_k| = \max_{1 \leq k \leq n} \delta_k = \frac{\varepsilon}{2} < \varepsilon.$$

Then $0 < \|T - S\| = \frac{\varepsilon}{2} < \varepsilon$. ■

Exercise 7F12

Suppose $T \in \mathcal{L}(V)$ is a positive operator. Show that $\|\sqrt{T}\| = \sqrt{\|T\|}$.

Solution. By 7.39, $\sqrt{T}$ is the unique positive square root of T, so T and $\sqrt{T}$ are self-adjoint by 7.38(b). By Exercise 7F2 and the nonnegativity of the eigenvalues of T and $\sqrt{T}$ by 7.38(b), we have

$$\|T\| = \max_{\lambda} \lambda, \quad \left\| \sqrt{T} \right\| = \max_{\mu} \mu,$$

where λ and μ range over all eigenvalues of T and $\sqrt{T}$, respectively.

By the proof of 7.39, for every eigenvalue λ of T, $\sqrt{\lambda}$ is an eigenvalue of $\sqrt{T}$. Moreover, if $\sqrt{T}v = \mu v$ for some nonzero $v \in V$, then $Tv = \mu^2 v$. Since $\mu \geq 0$, we have $\mu = \sqrt{\mu^2}$, and μ^2 is an eigenvalue of T since $v \neq 0$. Therefore, the eigenvalues of $\sqrt{T}$ are precisely the square roots of the eigenvalues of T, so

$$\left\| \sqrt{T} \right\| = \max_{\mu} \mu = \max_{\lambda} \sqrt{\lambda} = \sqrt{\max_{\lambda} \lambda} = \sqrt{\|T\|},$$

where the third equality follows from the fact that $\lambda \mapsto \sqrt{\lambda}$ is an increasing function on $[0, \infty)$. ■

Exercise 7F13

Suppose $S, T \in \mathcal{L}(V)$ are positive operators. Show that

$$\|S - T\| \leq \max\{\|S\|, \|T\|\} \leq \|S + T\|.$$

Solution. If $V = \{0\}$, then $S = T = 0$, and the inequality holds trivially. Now assume $V \neq \{0\}$. Since S and T are positive, the Cauchy–Schwarz inequality and 7.88(c) give

$$0 \leq \langle Sv, v \rangle \leq \|S\|, \quad 0 \leq \langle Tv, v \rangle \leq \|T\|$$

for every unit vector $v \in V$. By 7.38(b), S and T are self-adjoint, so $S - T$ is also self-adjoint by 7.5(a) and (b). By Exercise 7F2, there exists an eigenvalue λ of $S - T$ such that $\|S - T\| = |\lambda|$. Let $u \in V$ be a unit eigenvector corresponding to λ, so $(S - T)u = \lambda u$. Then

$$\lambda = \lambda \|u\|^2 = \langle (S - T)u, u \rangle = \langle Su, u \rangle - \langle Tu, u \rangle \in [-\|T\|, \|S\|].$$

Thus, $|\lambda| = \|S - T\| \leq \max\{\|S\|, \|T\|\}$.

By Exercise 7F2 again, there exists an eigenvalue μ of S such that $\|S\| = |\mu|$. By 7.38(b), $\mu \geq 0$, so $\|S\| = |\mu| = \mu$. Let $w \in V$ be a unit eigenvector corresponding to μ, so $Sw = \mu w$. Then $\langle Sw, w \rangle = \mu \|w\|^2 = \mu = \|S\|$. Consequently,

$$\|S\| \leq \|S\| + \langle Tw, w \rangle = \langle (S + T)w, w \rangle \leq \|S + T\|,$$

where the last inequality follows from the Cauchy–Schwarz inequality and 7.88(c). By symmetry, we also have $\|T\| \leq \|S + T\|$. Therefore, $\max\{\|S\|, \|T\|\} \leq \|S + T\|$, and the desired inequality is proved. ■

Exercise 7F14

Suppose U and W are subspaces of V such that $\|P_U - P_W\| < 1$. Prove that $\dim U = \dim W$.

Solution. Suppose $\dim U > \dim W$. By 6.51, 2.43, and 2.37,

$$\dim(U \cap W^{\perp}) = \dim U + \dim W^{\perp} - \dim(U + W^{\perp}) \geq \dim U - \dim W > 0.$$

Let $u \in U \cap W^{\perp}$ be a unit vector. Then $(P_U - P_W)u = P_U u - P_W u = u - 0 = u$ by 6.57(b) and (c). By 7.88(c), $\|P_U - P_W\| \geq \|(P_U - P_W)u\| = \|u\| = 1$, contradicting the assumption that $\|P_U - P_W\| < 1$. Therefore, our assumption that $\dim U > \dim W$ must be false. By symmetry, $\dim W > \dim U$ is also impossible. Hence, $\dim U = \dim W$. ■

Exercise 7F15

Define $T \in \mathcal{L}(\mathbb{F}^3)$ by

$$T(z_1, z_2, z_3) = (z_3, 2z_1, 3z_2).$$

Find (explicitly) a unitary operator $S \in \mathcal{L}(\mathbb{F}^3)$ such that $T = S\sqrt{T^*T}$.

Solution. For any $(z_1, z_2, z_3), (w_1, w_2, w_3) \in \mathbb{F}^3$,

$$\begin{aligned}\langle T(z_1, z_2, z_3), (w_1, w_2, w_3)\rangle &= \langle (z_3, 2z_1, 3z_2), (w_1, w_2, w_3)\rangle \\ &= z_3\overline{w_1} + 2z_1\overline{w_2} + 3z_2\overline{w_3} \\ &= \langle (z_1, z_2, z_3), (2w_2, 3w_3, w_1)\rangle.\end{aligned}$$

Therefore,

$$T^*(w_1, w_2, w_3) = (2w_2, 3w_3, w_1), \quad T^*T(z_1, z_2, z_3) = (4z_1, 9z_2, z_3).$$

Let $R \in \mathcal{L}(\mathbb{F}^3)$ be defined by $R(z_1, z_2, z_3) = (2z_1, 3z_2, z_3)$. Then $\mathcal{M}(R)$ with respect to the standard orthonormal basis e_1, e_2, e_3 of $\mathbb{F}^3$ is diagonal with nonnegative diagonal entries $2, 3, 1$. By 7.38(c), R is positive. Additionally, $R^2(z_1, z_2, z_3) = (4z_1, 9z_2, z_3) = T^*T(z_1, z_2, z_3)$, so $R^2 = T^*T$ and hence $R = \sqrt{T^*T}$ by 7.39. Moreover, R^{-1} exists and is given by

$$R^{-1}(z_1, z_2, z_3) = \left(\frac{z_1}{2}, \frac{z_2}{3}, z_3\right).$$

Define $S \in \mathcal{L}(\mathbb{F}^3)$ by $S = TR^{-1}$. Then

$$S(z_1, z_2, z_3) = T(R^{-1}(z_1, z_2, z_3)) = T\left(\frac{z_1}{2}, \frac{z_2}{3}, z_3\right) = (z_3, z_1, z_2).$$

Observe that $Se_1 = e_2$, $Se_2 = e_3$, and $Se_3 = e_1$. Since e_2, e_3, e_1 is an orthonormal basis of $\mathbb{F}^3$, S is unitary by 7.53(d). Because $S = TR^{-1}$, we have $T = SR = S\sqrt{T^*T}$. ■

Exercise 7F16

Suppose $S \in \mathcal{L}(V)$ is a positive invertible operator. Prove that there exists $\delta > 0$ such that T is a positive operator for every self-adjoint operator $T \in \mathcal{L}(V)$ with $\|S - T\| < \delta$.

Solution. If $V = \{0\}$, every operator on V is positive, so any $\delta > 0$ works. Now assume $V \neq \{0\}$. By 7.38(c), there exists an orthonormal basis $e_1, \ldots, e_n$ of V such that $Se_k = \lambda_k e_k$ with $\lambda_k \geq 0$ for all $k = 1, \ldots, n$. Then each λ_k is an eigenvalue of S because $e_k \neq 0$, and since S is invertible, $\lambda_k > 0$ for all $k = 1, \ldots, n$ by 5.7. Let

$$\delta = \min_{1 \leq k \leq n} \lambda_k > 0.$$

Observe that for any $v \in V$ and $k = 1, \ldots, n$, we have

$$\langle Sv, e_k\rangle = \langle v, Se_k\rangle = \overline{\lambda_k}\langle v, e_k\rangle = \lambda_k\langle v, e_k\rangle,$$

since S is self-adjoint and $\lambda_k \in \mathbb{R}$ for all k. By 6.30(b) and (c),

$$\langle Sv, v\rangle = \sum_{k=1}^{n} \langle Sv, e_k\rangle\overline{\langle v, e_k\rangle} = \sum_{k=1}^{n} \lambda_k |\langle v, e_k\rangle|^2 \geq \delta \sum_{k=1}^{n} |\langle v, e_k\rangle|^2 = \delta\|v\|^2.$$

Let $T \in \mathcal{L}(V)$ be self-adjoint with $\|S - T\| < \delta$. By the Cauchy–Schwarz inequality and 7.88(c),

$$\langle Tv, v\rangle = \langle Sv, v\rangle - \langle (S-T)v, v\rangle \geq \langle Sv, v\rangle - \|S-T\|\,\|v\|^2 \geq (\delta - \|S-T\|)\|v\|^2 \geq 0.$$

Since T is self-adjoint and $\langle Tv, v\rangle \geq 0$ for all $v \in V$, it follows that T is positive. ■

Exercise 7F17

Prove that if $u \in V$ and φ_u is the linear functional on V defined by the equation $\varphi_u(v) = \langle v, u\rangle$, then $\|\varphi_u\| = \|u\|$.

Axler Note. *Here we are thinking of the scalar field $\mathbb{F}$ as an inner product space with $\langle \alpha, \beta\rangle = \alpha\bar{\beta}$ for all $\alpha, \beta \in \mathbb{F}$. Thus, $\|\varphi_u\|$ means the norm of φ_u as a linear map from V to $\mathbb{F}$.*

Solution. Observe that $\|\alpha\| = \sqrt{\langle \alpha, \alpha\rangle} = \sqrt{\alpha\bar{\alpha}} = |\alpha|$ for all $\alpha \in \mathbb{F}$. For any $v \in V$, the Cauchy–Schwarz inequality gives

$$|\varphi_u(v)| = |\langle v, u\rangle| \leq \|v\|\,\|u\|.$$

By 7.88(c), $\|\varphi_u\| \leq \|u\|$.

If $u = 0$, then $0 \leq \|\varphi_u\| \leq \|u\| = 0$ by 7.87(a). For $u \neq 0$, consider $w = \frac{u}{\|u\|}$. Since $\|w\| = 1$, 7.88(b) gives $\|\varphi_u\| \geq |\varphi_u(w)|$, so

$$\|\varphi_u\| \geq |\varphi_u(w)| = |\langle w, u\rangle| = \left|\left\langle \frac{u}{\|u\|}, u\right\rangle\right| = \frac{\|u\|^2}{\|u\|} = \|u\|.$$

Hence, $\|\varphi_u\| = \|u\|$. ■

Exercise 7F18

Suppose $e_1, \ldots, e_n$ is an orthonormal basis of V and $T \in \mathcal{L}(V, W)$.

(a) Prove that $\max\{\|Te_1\|, \ldots, \|Te_n\|\} \leq \|T\| \leq (\|Te_1\|^2 + \cdots + \|Te_n\|^2)^{1/2}$.

(b) Prove that $\|T\| = (\|Te_1\|^2 + \cdots + \|Te_n\|^2)^{1/2}$ if and only if $\dim \operatorname{range} T \leq 1$.

Axler Note. *Here $e_1, \ldots, e_n$ is an arbitrary orthonormal basis of V, not necessarily connected with a singular value decomposition of T. If $s_1, \ldots, s_n$ is the list of singular values of T, then the right side of the inequality above equals $(s_1^2 + \cdots + s_n^2)^{1/2}$, as was shown in* Exercise 7E11(a).

Solution. Let $s_1 \geq \cdots \geq s_n$ be the singular values of T.

(a) Observe that $\max\{\|Te_1\|, \ldots, \|Te_n\|\} \leq \|T\|$ by 7.88(b) applied to each e_k. By Exercise 7E11(a) and 7.88(a),

$$\sum_{k=1}^{n} \|Te_k\|^2 = \sum_{k=1}^{n} s_k^2 \geq s_1^2 = \|T\|^2.$$

Taking square roots gives the desired inequality.

(b) Note that $\|T\| = (\sum_{k=1}^{n} \|Te_k\|^2)^{1/2}$ if and only if $\|T\|^2 = \sum_{k=1}^{n} \|Te_k\|^2$, since both sides are nonnegative.

$$\begin{aligned}
\|T\|^2 = \sum_{k=1}^{n} \|Te_k\|^2 &\iff s_1^2 = \sum_{k=1}^{n} s_k^2 \\
&\iff \sum_{k=2}^{n} s_k^2 = 0 \\
&\iff s_2 = \cdots = s_n = 0 \\
&\iff T \text{ has at most one positive singular value} \\
&\iff \dim \operatorname{range} T \le 1,
\end{aligned}$$

where the first equivalence follows from 7.88(a) and Exercise 7E11(a); the third equivalence from $s_k^2 \ge 0$ for all k; the fourth equivalence from the decreasing order of the singular values; and the fifth equivalence from 7.68(b). ■

Exercise 7F19

Prove that if $T \in \mathcal{L}(V, W)$, then $\|T^*T\| = \|T\|^2$.

Axler Note. *This formula for $\|T^*T\|$ leads to the important subject of C^*-algebras.*

Manual Remark. *Alternatively, if $T \neq 0$ and*

$$Tv = \sum_{k=1}^{m} s_k \langle v, e_k \rangle f_k$$

is a singular value decomposition of T by 7.70, then 7.75 gives

$$T^*Tv = \sum_{k=1}^{m} s_k^2 \langle v, e_k \rangle e_k,$$

so Exercise 7E8(c) *and 7.88(a) give* $\|T^*T\| = s_1^2 = \|T\|^2$.

Solution.

$\boxed{\le}$ By 7.88(b), there exists $v \in V$ with $\|v\| = 1$ such that $\|Tv\| = \|T\|$. By the definition of the adjoint, the Cauchy–Schwarz inequality, and 7.88(c), we have

$$\|T\|^2 = \|Tv\|^2 = \langle Tv, Tv \rangle = \langle v, T^*Tv \rangle \le \|v\| \, \|T^*Tv\| \le \|T^*T\| \, \|v\|^2 = \|T^*T\|.$$

$\boxed{\ge}$ By Exercise 7F5 and 7.91, $\|T^*T\| \le \|T^*\| \, \|T\| = \|T\|^2$. ■

Exercise 7F20

Suppose $T \in \mathcal{L}(V)$ is normal. Prove that $\|T^k\| = \|T\|^k$ for every positive integer k.

Solution. Let $n = \dim V$ and let $S = T^*T$. We first show that $(T^k)^*T^k = S^k$ for all positive integers k. The base case $k = 1$ is trivial since $(T^1)^*T^1 = T^*T = S$. Now suppose $(T^{k-1})^*T^{k-1} = S^{k-1}$ for $k \ge 2$. Since T is normal, S commutes with T, and hence with T^{k-1}. By 7.5(d) applied to T and T^{k-1} and the

induction hypothesis, we have

$$(T^k)^*T^k = (TT^{k-1})^*T^k = (T^{k-1})^*(T^*T)T^{k-1} = (T^{k-1})^*ST^{k-1} = (T^{k-1})^*T^{k-1}S = S^{k-1}S = S^k,$$

so $(T^k)^*T^k = S^k$ for all positive integers k.

Observe that S is positive by 7.64(a), so by 7.38(c), $\mathcal{M}(S)$ is diagonal with entries $\lambda_1, \dots, \lambda_n \geq 0$ with respect to some orthonormal basis of V. By 3.43, $\mathcal{M}(S^k)$ is also diagonal with entries $\lambda_1^k, \dots, \lambda_n^k$ with respect to the same orthonormal basis of V, which are exactly the eigenvalues of S^k by 5.41, so S^k is positive by 7.38(c). By Exercise 7F2 applied to S^k and S, the nonnegativity of the eigenvalues of S^k, and the fact that $x \mapsto x^k$ is increasing on $[0, \infty)$, we have

$$\|S^k\| = \max_{1 \leq j \leq n} \lambda_j^k = \left(\max_{1 \leq j \leq n} \lambda_j\right)^k = \|S\|^k.$$

The above computation and Exercise 7F19 applied to T^k and T then give $\|T^k\|^2 = \|S\|^k = \|T\|^{2k}$, and taking the nonnegative square roots of both sides yields $\|T^k\| = \|T\|^k$. ■

Exercise 7F21

Suppose $\dim V > 1$ and $\dim W > 1$. Prove that the norm on $\mathcal{L}(V, W)$ does not come from an inner product. In other words, prove that there does not exist an inner product on $\mathcal{L}(V, W)$ such that

$$\max\{\|Tv\| \mid v \in V \text{ and } \|v\| \leq 1\} = \sqrt{\langle T, T \rangle}$$

for all $T \in \mathcal{L}(V, W)$.

Solution. Suppose the norm on $\mathcal{L}(V, W)$ comes from an inner product. By 6.35, there exist orthonormal bases $e_1, \dots, e_{\dim V}$ of V and $f_1, \dots, f_{\dim W}$ of W. Since $\dim V > 1$ and $\dim W > 1$, we may take the orthonormal lists e_1, e_2 in V and f_1, f_2 in W. Consider the maps S and R defined by

$$Sv = \langle v, e_1 \rangle f_1, \quad Rv = \langle v, e_2 \rangle f_2.$$

Note that $S, R \in \mathcal{L}(V, W)$ by 6.6(a). By the Cauchy–Schwarz inequality,

$$\|Sv\| = |\langle v, e_1 \rangle| \, \|f_1\| \leq \|v\| \leq 1$$

for $\|v\| \leq 1$. Since $\|Se_1\| = \|f_1\| = 1$, we have $\|S\| = 1$. Similarly, $\|R\| = 1$, using e_2 and f_2. By 6.24 and Bessel's inequality, we have for any $v \in V$ that

$$\|(S \pm R)v\|^2 = |\langle v, e_1 \rangle|^2 + |\langle v, e_2 \rangle|^2 \leq \|v\|^2.$$

It follows that $\|(S \pm R)v\| \leq \|v\| \leq 1$ for all $v \in V$ with $\|v\| \leq 1$, so $\|S \pm R\| \leq 1$. Because $(S \pm R)e_1 = f_1$, it follows that $\|S + R\| \geq 1$ and $\|S - R\| \geq 1$. Hence, $\|S + R\| = \|S - R\| = 1$. Consequently,

$$\|S + R\|^2 + \|S - R\|^2 = 1^2 + 1^2 = 2$$

but

$$2\|S\|^2 + 2\|R\|^2 = 2 \cdot 1^2 + 2 \cdot 1^2 = 4.$$

The parallelogram equality requires that $\|S + R\|^2 + \|S - R\|^2 = 2\|S\|^2 + 2\|R\|^2$, which is not the case here. Therefore, the norm on $\mathcal{L}(V, W)$ does not come from an inner product. ■

Exercise 7F22

Suppose $T \in \mathcal{L}(V, W)$. Let $n = \dim V$ and let $s_1 \geq \cdots \geq s_n$ denote the singular values of T. Prove that if $1 \leq k \leq n$, then

$$\min\{\|T|_U\| \mid U \text{ is a subspace of } V \text{ with } \dim U = k\} = s_{n-k+1}.$$

Solution. Let m be the number of positive singular values of T, so $s_{m+1} = \cdots = s_n = 0$. By 7.70, there exist orthonormal lists $e_1, \ldots, e_m$ in V and $f_1, \ldots, f_m$ in W such that

$$Tv = \sum_{j=1}^{m} s_j \langle v, e_j \rangle f_j$$

for all $v \in V$. Extend $e_1, \ldots, e_m$ to an orthonormal basis $e_1, \ldots, e_n$ of V by 6.35 if $m = 0$, or by 6.36 if $m > 0$. By 6.24,

$$\|Tv\|^2 = \sum_{j=1}^{m} s_j^2 |\langle v, e_j \rangle|^2 + \sum_{j=m+1}^{n} 0 \cdot |\langle v, e_j \rangle|^2 = \sum_{j=1}^{n} s_j^2 |\langle v, e_j \rangle|^2.$$

$\boxed{\leq}$ Consider the subspace $X = \operatorname{span}(e_{n-k+1}, \ldots, e_n)$ of V. By 6.25, $\dim X = k$. For every $x \in X$ and $j \leq n - k$, we have $\langle x, e_j \rangle = 0$ by the orthonormality of $e_1, \ldots, e_n$. By the above calculation,

$$\|Tx\|^2 = \sum_{j=n-k+1}^{n} s_j^2 |\langle x, e_j \rangle|^2 \leq s_{n-k+1}^2 \sum_{j=n-k+1}^{n} |\langle x, e_j \rangle|^2 = s_{n-k+1}^2 \|x\|^2,$$

where the last equality follows from 6.30(b). By 7.88(c), $\|T|_X\| \leq s_{n-k+1}$.
$\boxed{\geq}$ Let U be any subspace of V with $\dim U = k$. Let $Y = \operatorname{span}(e_1, \ldots, e_{n-k+1})$, so $\dim Y = n - k + 1$ by 6.25. By 2.37, $\dim(U + Y) \leq n$. By 2.43,

$$\dim(U \cap Y) = \dim U + \dim Y - \dim(U + Y) \geq k + (n - k + 1) - n = 1.$$

Let $u \in U \cap Y$ be nonzero. For all $j > n - k + 1$, we have $\langle u, e_j \rangle = 0$ by the orthonormality of $e_1, \ldots, e_n$. By the above calculation,

$$\|Tu\|^2 = \sum_{j=1}^{n-k+1} s_j^2 |\langle u, e_j \rangle|^2 \geq s_{n-k+1}^2 \sum_{j=1}^{n-k+1} |\langle u, e_j \rangle|^2 = s_{n-k+1}^2 \|u\|^2,$$

where the last equality follows from 6.30(b). Taking square roots yields $\|Tu\| \geq s_{n-k+1}\|u\|$. By 7.88(c), $\|T|_U\| \, \|u\| \geq \|Tu\| \geq s_{n-k+1}\|u\|$, so $\|T|_U\| \geq s_{n-k+1}$ by dividing both sides by $\|u\| > 0$ since $u \neq 0$. Since every U with $\dim U = k$ satisfies $\|T|_U\| \geq s_{n-k+1}$, and we have found an X with $\dim X = k$ such that $\|T|_X\| \leq s_{n-k+1}$, the minimum exists and is attained at X, and therefore equals s_{n-k+1}. ■

Exercise 7F23

Suppose $T \in \mathcal{L}(V, W)$. Show that T is uniformly continuous with respect to the metrics on V and W that arise from the norms on those spaces (see Exercise 6B23).
Manual Remark. *For reference, the metrics on V and W that arise from the norms are $d_V(v_1, v_2) = \|v_1 - v_2\|$ and $d_W(w_1, w_2) = \|w_1 - w_2\|$. With these metrics, uniform continuity of T means that for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $v_1, v_2 \in V$, if $\|v_1 - v_2\| < \delta$, then $\|Tv_1 - Tv_2\| < \varepsilon$.*

Solution. Let $\varepsilon > 0$ be given. Define

$$\delta = \frac{\varepsilon}{\|T\| + 1}.$$

Note that $\delta > 0$ since $\|T\| \geq 0$ by 7.87(a). Suppose $v_1, v_2 \in V$ such that $\|v_1 - v_2\| < \delta$. By 7.88(c),

$$\|Tv_1 - Tv_2\| = \|T(v_1 - v_2)\| \leq \|T\| \, \|v_1 - v_2\| \leq \|T\|\delta = \|T\| \frac{\varepsilon}{\|T\| + 1} < \varepsilon.$$

Hence, T is uniformly continuous. ■

Exercise 7F24

Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

$$\|T^{-1}\| = \|T\|^{-1} \iff \frac{T}{\|T\|} \text{ is a unitary operator.}$$

Solution. Since T is invertible, $T \neq 0$ so $\|T\| \neq 0$ by 7.87(b). Let $S = \frac{T}{\|T\|}$, so $S^{-1} = \|T\|T^{-1}$. By 7.87(c), $\|S\| = 1$.
$\boxed{\Rightarrow}$ Suppose $\|T^{-1}\| = \|T\|^{-1}$. By 7.87(c), $\|S^{-1}\| = \|T^{-1}\|\|T\| = 1$. Then for every $v \in V$, we have by 7.88(c) that

$$\|v\| = \|S^{-1}Sv\| \leq \|S^{-1}\| \, \|Sv\| = \|Sv\| \leq \|S\| \, \|v\| = \|v\|,$$

so $\|Sv\| = \|v\|$. Therefore, S is an invertible isometry, so S is unitary.
$\boxed{\Leftarrow}$ Suppose S is unitary. Then $S^{-1} = S^*$ by 7.53(c), and hence is unitary by 7.53(f). Thus, $\|S^{-1}\| = 1$ by 7.88(b), so $\|T\| \, \|T^{-1}\| = \|S^{-1}\| = 1$ by 7.87(c). Hence, $\|T^{-1}\| = \|T\|^{-1}$. ■

Exercise 7F25

Fix $u, x \in V$ with $u \neq 0$. Define $T \in \mathcal{L}(V)$ by $Tv = \langle v, u\rangle x$ for every $v \in V$. Prove that

$$\sqrt{T^*T}\, v = \frac{\|x\|}{\|u\|}\langle v, u\rangle u$$

for every $v \in V$.

Solution. Observe that $\langle Tv, w\rangle = \langle \langle v, u\rangle x, w\rangle = \langle v, u\rangle\langle x, w\rangle$ for all $v, w \in V$. Then $\langle v, u\rangle\langle x, w\rangle = \langle v, \overline{\langle x, w\rangle}u\rangle = \langle v, \langle w, x\rangle u\rangle$ for all $v, w \in V$, so $T^*w = \langle w, x\rangle u$ for all $w \in V$. Therefore,

$$T^*Tv = T^*(\langle v, u\rangle x) = \langle v, u\rangle T^*x = \langle v, u\rangle\langle x, x\rangle u = \|x\|^2\langle v, u\rangle u$$

for all $v \in V$. Define $R \in \mathcal{L}(V)$ by $Rv = \frac{\|x\|}{\|u\|}\langle v, u\rangle u$ for all $v \in V$. Since $\frac{\|x\|}{\|u\|} \geq 0$, $\langle Rv, w\rangle = \frac{\|x\|}{\|u\|}\langle v, u\rangle\langle u, w\rangle = \langle v, Rw\rangle$ for all $v, w \in V$, so R is self-adjoint. Moreover,

$$\langle Rv, v\rangle = \frac{\|x\|}{\|u\|}\langle v, u\rangle\langle u, v\rangle = \frac{\|x\|}{\|u\|}|\langle v, u\rangle|^2 \geq 0,$$

so R is positive. Finally,

$$R^2v = R\left(\frac{\|x\|}{\|u\|}\langle v, u\rangle u\right) = \frac{\|x\|}{\|u\|}\left\langle \frac{\|x\|}{\|u\|}\langle v, u\rangle u, u\right\rangle u = \frac{\|x\|^2}{\|u\|^2}\langle v, u\rangle\langle u, u\rangle u = \|x\|^2\langle v, u\rangle u = T^*Tv$$

for all $v \in V$. Then R is a positive square root of T^*T, so $R = \sqrt{T^*T}$ by 7.39. Hence,

$$\sqrt{T^*T}\, v = Rv = \frac{\|x\|}{\|u\|}\langle v, u\rangle u$$

for all $v \in V$. ■

Exercise 7F26

Suppose $T \in \mathcal{L}(V)$. Prove that T is invertible if and only if there exists a unique unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$.

Solution. Since $\sqrt{T^*T}$ is positive, it is self-adjoint. Then for any $v \in V$, we have

$$\left\| \sqrt{T^*T}\, v \right\|^2 = \langle T^*Tv, v\rangle = \|Tv\|^2.$$

Hence, $\operatorname{null} \sqrt{T^*T} = \operatorname{null} T$. By the polar decomposition of T, the existence of such S is guaranteed. As such, it remains to show that S is unique if and only if T is invertible.

$\boxed{\Rightarrow}$ Suppose T is invertible. By the above, $\operatorname{null} \sqrt{T^*T} = \operatorname{null} T = \{0\}$, so $\sqrt{T^*T}$ is injective, and hence invertible by 3.65. Moreover, if $S_1, S_2 \in \mathcal{L}(V)$ are unitary operators such that $T = S_1\sqrt{T^*T} = S_2\sqrt{T^*T}$, then right-multiplying by $(\sqrt{T^*T})^{-1}$ gives $S_1 = S_2$, so S is unique.

$\boxed{\Leftarrow}$ Suppose T is not invertible. By 3.65, $\operatorname{null} T \neq \{0\}$. By the polar decomposition of T, there exists a unitary operator S such that $T = S\sqrt{T^*T}$. By 6.35, there exists an orthonormal basis $e_1, \ldots, e_m$ of $\operatorname{null} T$, where $m \geq 1$. By 6.36, we may extend this to an orthonormal basis $e_1, \ldots, e_n$ of V. By the linear map lemma, there exists $R \in \mathcal{L}(V)$ such that $Re_j = -e_j$ for $j \leq m$ and $Re_j = e_j$ for $j > m$. Because $Re_1, \ldots, Re_n$ is an orthonormal basis of V, R is unitary by 7.53(d). Moreover, self-adjointness of $\sqrt{T^*T}$ gives

$$\operatorname{range} \sqrt{T^*T} = (\operatorname{null} \sqrt{T^*T})^\perp = (\operatorname{null} T)^\perp$$

by 7.6(b). Now, if $u \in (\operatorname{null} T)^\perp$, then $\langle u, e_j\rangle = 0$ for $j \leq m$. By 6.30(a), we may write

$$u = \sum_{j=m+1}^{n} \langle u, e_j\rangle e_j,$$

so $Ru = u$. Consequently, $R\sqrt{T^*T} = \sqrt{T^*T}$, so $SR\sqrt{T^*T} = S\sqrt{T^*T} = T$. By 7.5(d) and 7.53(b),

$$(SR)^*(SR) = R^*S^*SR = I, \quad (SR)(SR)^* = SRR^*S^* = I.$$

Hence, SR is unitary. Lastly, injectivity of S shows that $Se_1 \neq 0$, so $SRe_1 = -Se_1 \neq Se_1$. Therefore, S and SR are distinct unitary operators such that $T = S\sqrt{T^*T} = SR\sqrt{T^*T}$, and S is not unique. ■

Exercise 7F27

Suppose $T \in \mathcal{L}(V)$ and $s_1, \ldots, s_n$ are the singular values of T. Let $e_1, \ldots, e_n$ and $f_1, \ldots, f_n$ be orthonormal bases of V such that

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n$$

for all $v \in V$. Define $S \in \mathcal{L}(V)$ by

$$Sv = \langle v, e_1\rangle f_1 + \cdots + \langle v, e_n\rangle f_n.$$

(a) Show that S is unitary and $\|T - S\| = \max\{|s_1 - 1|, \ldots, |s_n - 1|\}$.
(b) Show that if $E \in \mathcal{L}(V)$ is unitary, then $\|T - E\| \geq \|T - S\|$.

Axler Note. *This exercise finds a unitary operator S that is as close as possible (among the unitary operators) to a given operator T.*

Solution.

(a) Since $Se_j = f_j$ for all j, S is unitary by 7.53(d). By assumption,

$$(T - S)v = \sum_{j=1}^{n}(s_j - 1)\langle v, e_j\rangle f_j$$

for any $v \in V$. Define

$$M = \max_{1 \leq j \leq n} |s_j - 1|.$$

By 6.24 and 6.30(b),

$$\|(T - S)v\|^2 = \sum_{j=1}^{n} |s_j - 1|^2 |\langle v, e_j\rangle|^2 \leq M^2 \sum_{j=1}^{n} |\langle v, e_j\rangle|^2 = M^2\|v\|^2$$

for all $v \in V$. Taking square roots and applying 7.88(c) gives $\|T - S\| \leq M$. Choose k such that $|s_k - 1| = M$. Since $\|(T - S)e_k\| = |s_k - 1| = M$ and $\|e_k\| = 1$, 7.88(b) gives $\|T - S\| \geq M$. Therefore, $\|T - S\| = M$.

(b) Suppose $E \in \mathcal{L}(V)$ is unitary. Then it is an isometry, so $\|Ee_j\| = \|e_j\| = 1$ for all j. Moreover, since $Te_j = s_j f_j$ and $s_j \geq 0$ for all j, we have $\|Te_j\| = s_j$. Because $\|e_j\| = 1$, 7.88(b) and the reverse triangle inequality give

$$\|T - E\| \geq \|(T - E)e_j\| \geq \big|\|Te_j\| - \|Ee_j\|\big| = |s_j - 1|.$$

Since this holds for all j, taking the maximum over j gives

$$\|T - E\| \geq \max_{1 \leq j \leq n} |s_j - 1| = \|T - S\|.$$ ■

Exercise 7F28

Suppose $T \in \mathcal{L}(V)$. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = \sqrt{TT^*}\, S$.

Solution. By the polar decomposition of T^*, there exists a unitary operator $U \in \mathcal{L}(V)$ such that $T^* = U\sqrt{(T^*)^*T^*}$. By 7.5(c), $(T^*)^* = T$, so $T^* = U\sqrt{TT^*}$. Since $\sqrt{TT^*}$ is positive, it is self-adjoint, so $\sqrt{TT^*} = (\sqrt{TT^*})^*$. By 7.5(c), (d), and taking adjoints, $T = (U\sqrt{TT^*})^* = \sqrt{TT^*}\,U^*$. By 7.53(f), U^* is unitary since U is unitary. Setting $S = U^*$ gives the desired result. ■

Exercise 7F29

Suppose $T \in \mathcal{L}(V)$.

(a) Use the polar decomposition to show that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $TT^* = ST^*TS^*$.

(b) Show how (a) implies that T and T^* have the same singular values.

Solution.

(a) By the polar decomposition of T, there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$. By 7.5(d), $T^* = (\sqrt{T^*T})^*S^*$. Since $\sqrt{T^*T}$ is positive, it is self-adjoint, so $(\sqrt{T^*T})^* = \sqrt{T^*T}$, and hence $T^* = \sqrt{T^*T}\,S^*$. By the definition of the square root,

$$TT^* = S\sqrt{T^*T}\left(\sqrt{T^*T}\,S^*\right) = ST^*TS^*.$$

(b) Since T^*T is positive, 7.38(c) gives an orthonormal basis $e_1, \dots, e_n$ of V and nonnegative numbers $\mu_1, \dots, \mu_n$ such that $T^*Te_k = \mu_k e_k$ for each k. By 7.53(b), $S^*S = I$ since S is unitary. Therefore,

$$TT^*(Se_k) = ST^*TS^*(Se_k) = ST^*Te_k = S(\mu_k e_k) = \mu_k(Se_k).$$

Since S is invertible, $Se_1, \dots, Se_n$ is also a basis of V. By Lemma 1 applied to T with the basis $e_1, \dots, e_n$, the singular values of T are $\sqrt{\mu_1}, \dots, \sqrt{\mu_n}$ sorted into decreasing order. By 7.5(c), $(T^*)^*T^* = TT^*$, so $(T^*)^*T^*(Se_k) = TT^*(Se_k) = \mu_k(Se_k)$ for each k. By Lemma 1 applied to T^* with the basis $Se_1, \dots, Se_n$, the singular values of T^* are $\sqrt{\mu_1}, \dots, \sqrt{\mu_n}$ sorted into decreasing order, which are the same as those of T. ■

Exercise 7F30

Suppose $T \in \mathcal{L}(V)$, $S \in \mathcal{L}(V)$ is a unitary operator, and $R \in \mathcal{L}(V)$ is a positive operator such that $T = SR$. Prove that $R = \sqrt{T^*T}$.

Axler Note. *This exercise shows that if we write T as the product of a unitary operator and a positive operator (as in the polar decomposition 7.93), then the positive operator equals $\sqrt{T^*T}$.*

Solution. Since R is positive, it is self-adjoint. By 7.5(d) and 7.53(b), $T^*T = RS^*SR = R^2$. Then R is a positive square root of T^*T, so $R = \sqrt{T^*T}$ by 7.39. ■

Exercise 7F31

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$ and such that S and $\sqrt{T^*T}$ both have diagonal matrices with respect to the same orthonormal basis of V.

Solution. By the complex spectral theorem, there exists an orthonormal basis $e_1, \dots, e_n$ of V consisting of eigenvectors of T, and scalars $\lambda_1, \dots, \lambda_n$ such that $Te_k = \lambda_k e_k$ for each k. By 7.21(e),

$$T^*Te_k = \lambda_k T^* e_k = \lambda_k \overline{\lambda_k} e_k = |\lambda_k|^2 e_k.$$

By the linear map lemma, there exists $R \in \mathcal{L}(V)$ such that $Re_k = |\lambda_k| e_k$ for each k. Then $\mathcal{M}(R, (e_1, \dots, e_n))$ is diagonal with entries $|\lambda_1|, \dots, |\lambda_n|$, so R is positive by 7.38(c) and (a). Moreover, $R^2 e_k = |\lambda_k|^2 e_k = T^*Te_k$ for each k, so $R^2 = T^*T$ by the linear map lemma. Since R is a positive square root of T^*T, we have $R = \sqrt{T^*T}$ by 7.39. By the linear map lemma again, there exists $S \in \mathcal{L}(V)$ such that $Se_k = \mu_k e_k$, where

$$\mu_k = \begin{cases} \frac{\lambda_k}{|\lambda_k|}, & \lambda_k \neq 0, \\ 1, & \lambda_k = 0. \end{cases}$$

Since each $Se_k = \mu_k e_k$ with $|\mu_k| = 1$, it follows that $Se_1, \dots, Se_n$ is an orthonormal basis of V, so S is unitary by 7.53(d). Moreover, $Te_k = \lambda_k e_k = \mu_k |\lambda_k| e_k = SRe_k$ for each k, so $T = SR = S\sqrt{T^*T}$ by the linear map lemma. It now follows that $\mathcal{M}(S, (e_1, \dots, e_n))$ is diagonal with entries $\mu_1, \dots, \mu_n$, and $\mathcal{M}(\sqrt{T^*T}, (e_1, \dots, e_n))$ is diagonal with entries $|\lambda_1|, \dots, |\lambda_n|$, so S and $\sqrt{T^*T}$ both have diagonal matrices with respect to the same orthonormal basis $e_1, \dots, e_n$. ■

Exercise 7F32

Suppose that $T \in \mathcal{L}(V, W)$ and $T \neq 0$. Let $s_1, \dots, s_m$ denote the positive singular values of T. Show that there exists an orthonormal basis $e_1, \dots, e_m$ of $(\operatorname{null} T)^\perp$ such that

$$T\left(E\left(\frac{e_1}{s_1}, \dots, \frac{e_m}{s_m}\right)\right)$$

equals the ball in $\operatorname{range} T$ of radius 1 centered at 0.

Solution. Let $B = \{w \in \operatorname{range} T \mid \|w\| < 1\}$. Since $T \neq 0$, we have $\dim \operatorname{range} T = m \geq 1$ by 7.68(b). By the singular value decomposition, there exist orthonormal lists $e_1, \dots, e_m$ in V and $f_1, \dots, f_m$ in W such that

$$Tv = \sum_{k=1}^{m} s_k \langle v, e_k \rangle f_k$$

for all $v \in V$. Moreover, for any $u \in \operatorname{null} T$,

$$Tu = \sum_{k=1}^{m} s_k \langle u, e_k \rangle f_k = 0.$$

Since $f_1, \dots, f_m$ is linearly independent by 6.25 and $s_k > 0$ for all k, $\langle u, e_k \rangle = 0$ for all $k = 1, \dots, m$, so $e_k \in (\operatorname{null} T)^\perp$ for all k. By 6.51 and the fundamental theorem of linear maps, $\dim(\operatorname{null} T)^\perp = \dim \operatorname{range} T = m$, so $e_1, \dots, e_m$ form an orthonormal basis of $(\operatorname{null} T)^\perp$ by 6.28. By 7.96 in $(\operatorname{null} T)^\perp$,

$$E = E\left(\frac{e_1}{s_1}, \dots, \frac{e_m}{s_m}\right) = \left\{ v \in (\operatorname{null} T)^\perp \;\middle|\; \sum_{k=1}^{m} s_k^2 |\langle v, e_k \rangle|^2 < 1 \right\}.$$

By 6.24, $\|Tv\|^2 = \sum_{k=1}^{m} s_k^2 |\langle v, e_k \rangle|^2$ for all $v \in V$, so

$$E = \{v \in (\operatorname{null} T)^\perp \mid \|Tv\| < 1\}.$$

We now show that $T(E) = B$.
$\boxed{\subseteq}$ This follows immediately from the definition of E.
$\boxed{\supseteq}$ Suppose $w \in B$. By 6.67, there exists $v \in (\operatorname{null} T)^{\perp}$ such that $Tv = w$. Then $\|Tv\| < 1$, so $v \in E$, and hence $w = Tv \in T(E)$. ■